

CLINICAL TECHNIQUE FOR DIRECT CLASS III, CLASS IV AND CLASS V RESTORATIONS

Class III Tooth Preparation for composite restorations:

The tooth preparation for a Class III direct composite restoration involves

- (1) Obtaining access to the defect (caries lesion, fracture)
- (2) Removing faulty structures (cariou tissue, defective dentin and enamel, defective restoration, base material)
- (3) Creating the convenience form for the restoration.

When a proximal surface of an anterior tooth is to be restored, and a choice between *facial* or *lingual* entry into the tooth is available, the lingual approach is always preferable unless

The advantages of restoring the proximal lesion using a *lingual* approach are:

1. The facial enamel is conserved for enhanced esthetics.
2. Shade matching of the composite is less critical.
3. Discoloration of the restoration is less visible.

Indications for a *facial* approach include:

1. The caries lesion is positioned facially, and facial access would significantly conserve the tooth structure.
2. Teeth are irregularly aligned, and facial access would significantly conserve the tooth structure.

Pre-clinical Operative Dentistry

3. An extensive caries lesion extends onto the facial surface.
 4. A faulty restoration that originally was placed from the facial approach needs to be replaced.
- ✓ Conventional cavity preparation (large caries, defective filling and root caries).
 - ✓ Modified cavity preparation (small to moderate lesion) figure 1.

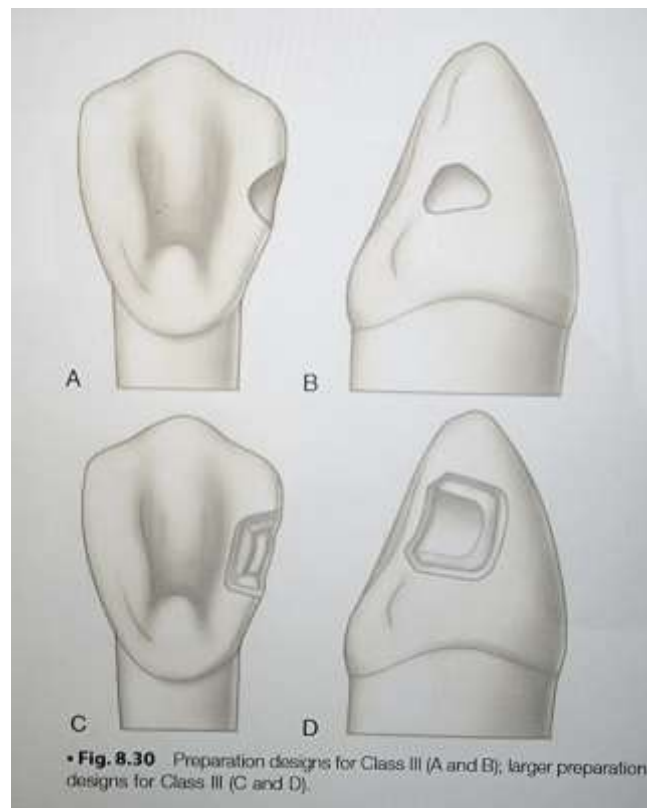


Figure 1: A&B Cl III cavity design for small caries, C&D Cl III cavity design for large lesion

Tooth preparation for Cl III composite restoration

The preparation is initiated from a lingual approach (if possible) by using a round carbide bur or diamond instrument of a size compatible with the extent of the lesion. The point of entry is located within the incisogingival dimension of the lesion or defect and as close to the adjacent tooth as possible without contacting it (Fig. 8.28A).

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The initial opening is made by holding the cutting instrument perpendicular to the enamel surface but at an entry angle that places the neck portion of the bur or diamond instrument as far into the embrasure (next to the adjacent tooth) as possible;

1. Using a No. ½, 1, 2 round bur prepare the outline form on the root surface
2. Extend the preparation into sound walls
3. Extend pulpally 0.75mm in depth
4. The gingival/cervical and incisal wall is perpendicular to the root surface (box like design)
5. A continuous groove retention can be prepared 0.25 mm (½ of diameter of bur) into dentin of the gingival and incisal walls with a ¼ round bur.
6. Larger preparations may require additional beveling of the accessible enamel walls to enhance retention by bonding.
7. Clean preparation and inspect the final preparation.

Lingual Access

1. Use a round bur No. 1/2, 1. 2 depending on the size of the caries to enlarge the opening sufficiently to allow for caries removal.
2. Extend external walls to sound tooth structure using a straight bur
3. Extend the gingival and incisal walls up to extent of caries or location of old restoration.

Unless necessary, DO NOT:

- Extend into the facial surface.

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- Extend subgingivally
4. Create an initial axial wall depth of 0.2mm into the dentin/DEJ (approximately 0.75 – 1.25mm in depth)
 5. Axial wall is convex, following the external contour of the tooth.
 6. Remove all remaining infected dentin, using a round bur or small spoon excavator.
 7. Remove unsupported enamel at the margins.
 8. If necessary, prepare extra retention means (grooves or coves) figure 2.
- Prepare it along the gingiva-axial line angle, and sometimes at the inciso-axial line angle 0.25 mm with a ¼ round bur.

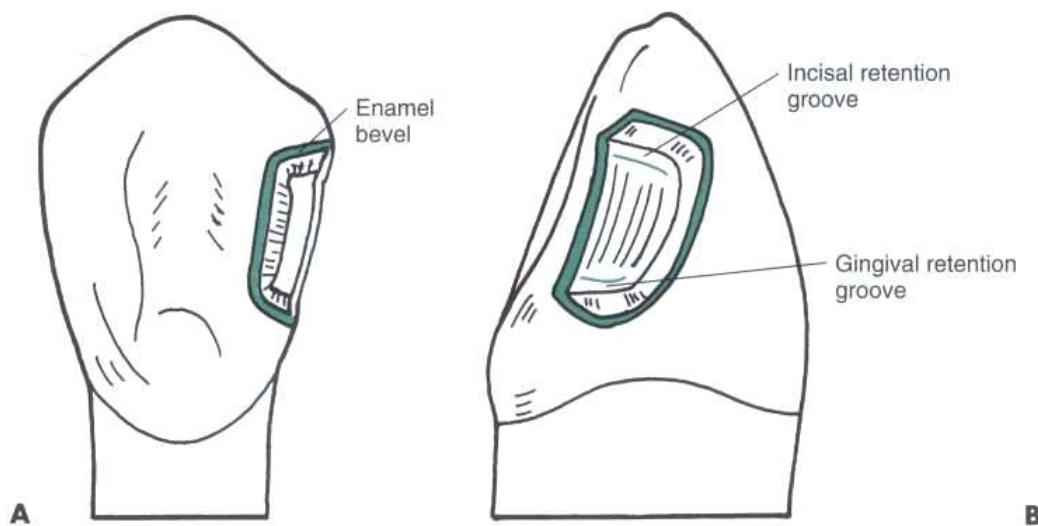


Figure 2: Extra-retention means (grooves and coves) in CI III cavity preparation.

9. Place cavosurface bevel or flare at the enamel except at the gingival margin area figure 3.
10. Use a flame shape or round bur resulting in a 45 degrees angle to the external tooth surface.

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11. Bevel width should be 0.25 to 0.5mm.
12. Clean the preparation of any debris and inspect final preparation.

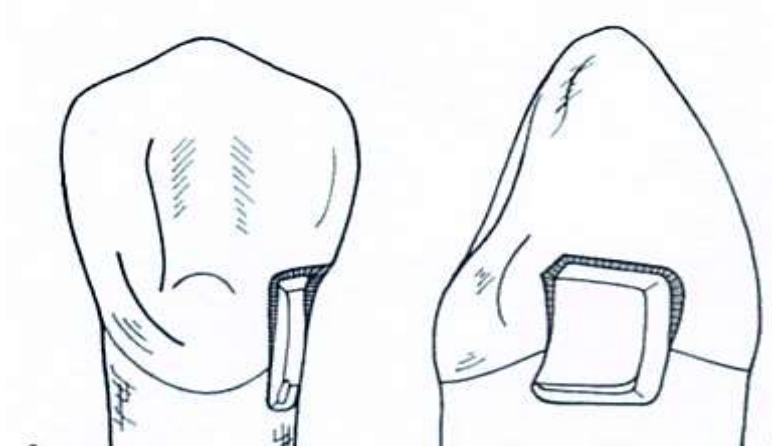


Figure 3: Bevel placement in Cl III.

Facial Access

- Same stages and steps are followed
- procedure is simplified because of easy access

Modified Class III

- Most used type of cavity preparation.
 - Indicated for small and moderate lesions or faults figure 1.
 - Designed to be as conservative as possible.
 - Preparation walls have no specific shapes or forms.
 - Preparation design appears to be scooped or concave
1. Use a 1/2, 1, 2 round bur, point of entry is within the inciso-gingival dimension of the lesion, perpendicular to the enamel surface.
 2. Remove all remaining caries or defect.

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3. No attempt is made to create a uniform axial wall.
4. Place cavosurface bevel or flare at the enamel except at the gingival margin area.
5. Use a flame shape or round bur resulting in a 45 degrees angle to the external tooth surface.
6. Bevel width should be 0.25 to 0.5mm.
7. Clean the preparation of any debris and inspect final preparation.

Class IV Tooth Preparation

- Preparation is similar to Class III except that the preparation for Cl IV is extended to the Incisal angles.
- Conventional cavity preparation (large caries, defective filling and root caries) figure 4 C&D.
- Modified cavity preparation (small to moderate lesion) figure 4 A&B.

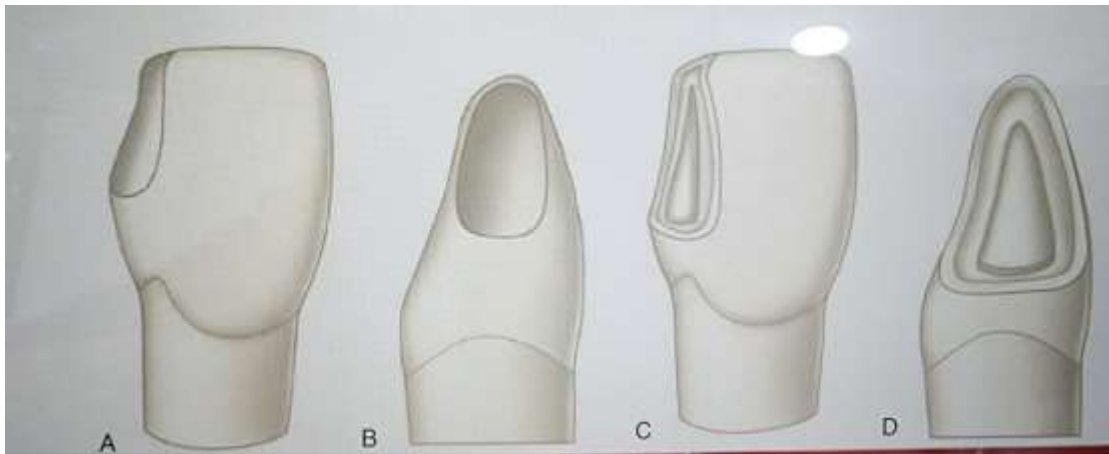


Figure 4: Cl IV cavity preparation.

- The tooth preparation for large Inciso-proximal areas requires more attention to the retention form than that for a small Class IV defect.
- If a large amount of tooth structure is missing and the restoration is in a high occlusal stress area, groove retention form may be indicated even when the preparation periphery is entirely in enamel.

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- Also, enamel bevels can be increased in width to provide greater surface area for etching, resulting in a stronger bond between the composite and the tooth, and potentially a better esthetic result.

Class V Tooth Preparation

- Have a multifactorial etiology in which contributing factors include intrinsic and/or extrinsic erosion, abrasion, abfraction, and/or biocorrosion.
- Conventional cavity preparation (large caries, defective filling and root caries) figure 5 C&D.
- Modified cavity preparation (small to moderate lesion) figure 5 A&B.

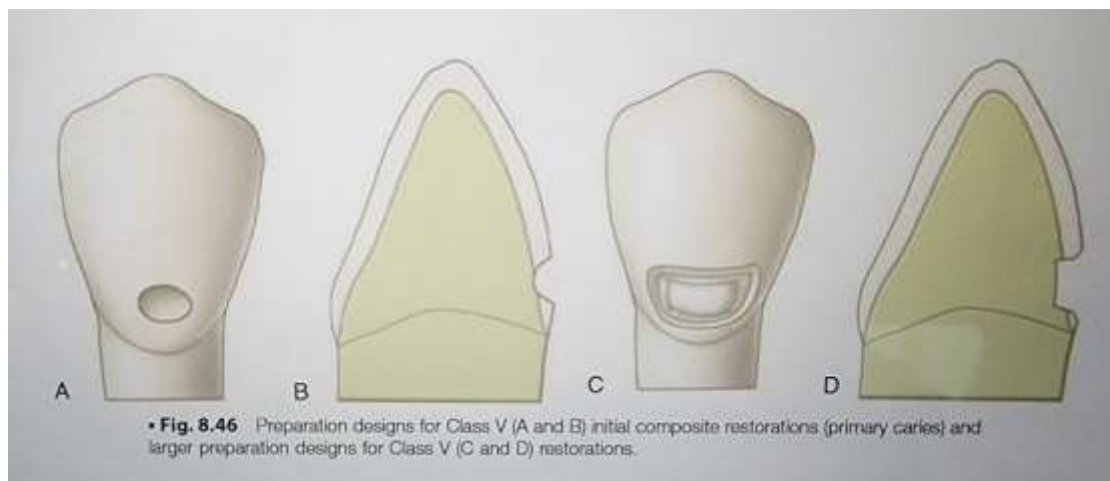


Figure 5: CI V cavity preparation.

Conventional

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- The feature of the preparation include a 90 degree cavosurface angle, uniform depth of the axial line angle, and sometimes, groove retention form.
 - Enamel bevels are typically used on the occlusal/incisal margins of the preparation, while at the cervical margin, bevels are usually not recommended because of the absence of enamel in this area.
1. Use a tapered fissure (No. 700, 701, or 271) or No.1 or 2 round bur.
 2. Make entry at 45 degrees angle to tooth surface, this should result to a 90 degree cavosurface.
 3. Axial depth is 0.75 mm to strength of preparation wall, strength of composite and placement of retention groove
 4. Axial should follow contour of the tooth.
 5. Extent of outline form is dictated by the carious lesion extent.
 6. Remove remaining carious lesion
 7. Prepare retention groove (similar to Class III preparation)
 8. Clean preparation

Modified Class V

- No effort to prepare a butt-joint
- No retention groove
- Lesion is scooped out
- Preparation has divergent wall
- Axial wall does not have uniform depth
- Prepare tooth with round or elliptical instrument
- Preparation is extended initially no deeper than 0.2 mm

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- No effort is made to prepare a 90 degree cavosurface margins.
- Infected enamel is removed with a round bur or excavator.